

Hu Lifan

+65 80999132 • lifan.hu@u.nus.edu • www.linkedin.com/in/anormalm/ • anormalm.com/

EDUCATION**National University of Singapore***Aug 2024 - Present**Bachelor of Engineering in **Computer Engineering****Second Major: Innovation and Design Program, Minor: Mathematics GPA:4.6***Shanghai Jiao Tong University (Summer School)***Jun 2025 - Jul 2025**Courses: Algebra, Statistics Inference*

PUBLICATIONS

Lifan Hu, "Learning Lie Group Generators From Trajectories," arXiv.org, April 4, 2025,
<https://arxiv.org/abs/2504.03220>.

Lifan Hu, "Linguine: A Natural-Language Programming Language with Formal Semantics and a Clean Compiler Pipeline," arXiv.org, June 10, 2025, <https://arxiv.org/abs/2506.08396>.

Lifan Hu, "GNN-Augmented RL for Fraud Detection in Decentralized Finance," CONF-SEML 2025(Published), April 14, 2025, <https://doi.org/10.54254/2755-2721/2025.22856>.

Lifan Hu, "Graph Evidence Is Not Text Evidence: Serialization Faithfulness as Structural Invariance in Graph-Grounded Language Models," *accepted as poster at EMNLP 2026 GroundLM Workshop as an Archival Short Paper*, 2026.

Lifan Hu, Chew Hoe Alfred Ang, Li Hong Idris Lim, "Quantized Vision-language models on Embedded GPUs: A Comparative Benchmark for Edge Multimodal Inference," *accepted as oral at the 19th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2026.

Lifan Hu *Anonymous Manuscript Under Review for AAAI 2026*, 2026.

INTERNSHIP EXPERIENCE**TikTok Pte. Ltd., Machine Learning Engineer****Singapore***Intern, TikTok BRIC**May 2026 -**Present*

- Developed graph-based integrity-risk discovery methods for large-scale live-streaming ecosystems using k-hop subgraphs, Leiden/Louvain community detection, Personalized PageRank, and structural anomaly signals.
- Built distributed Spark pipelines that process tens of millions of records in tens of minutes and generate risk-ranked accounts, transactions, and coordinated communities for downstream investigation.
- Designed graph anomaly-scoring methods under incomplete and weak supervision, achieving mid-0.8 ROC-AUC and approximately 3–4× top-decile lift in offline experiments.
- Analyzed coordinated, sparse, and adaptive harmful behaviors through community structure, temporal propagation, account relationships, and transaction-pattern features.
- Integrated graph algorithms with scalable batch-processing infrastructure for production-oriented risk discovery.
- Co-designed Case Agent workflow for internal use on monitoring live room fund risks and discovery.

Shanghai MAHLE Thermal Systems Co., Ltd., Digital Developer**Shanghai ,****CN***Intern, Finance, Controlling & IT**May 2025 - July**2025*

- Wrote a real-time multilingual meeting transcriber/translator using Vosk and MarianMT; fully offline-capable, deployed in constrained industrial environments with over 95% ASR accuracy in quiet speech.
- Trained a YOLOv5 model on blueprint .tif images, achieving over 85% precision in detecting diagrams, tables, and handwritten notes in manufacturing documents.
- Developed an OCR pipeline combining Tesseract and PaddleOCR to extract structured data from multi-format scanned documents with layout reconstruction, achieving over 98% text accuracy.
- Created an OCR-based screen-snipping data extractor for QM factory interfaces, integrating image hashing and keyboard automation to enable zero-integration telemetry, now operating on 90% of local factory interfaces.
- Deployed multiple lightweight Flask APIs to serve model inference, real-time OCR, and monitoring dashboards across local networked systems.

RESEARCH EXPERIENCE

1. GNN + MARL for DeFi Fraud Detection

Jan 2025 - May 2025

Remote Researcher, Supervised by Prof. Pietro Liò, Cambridge University

- Built a GNN-augmented multi-agent PPO system in a custom PettingZoo ParallelEnv to detect fraudulent behaviours in DeFi transaction graphs; validated against 50K+ Ethereum records
- Benchmarked hybrid GNN-RL-GAN model against RL-only and traditional ML with 50k Ethereum records

2. On-Device Vision-Language Assistant for the Visually Impaired (Merit Award)

Aug 2025 - Feb 2026

Undergraduate Researcher, Supervised by A/Prof. Lim Li Hong Idris, National University of Singapore

- Designed and implemented an edge-based multimodal perception pipeline integrating quantized Vision-Language Models (VILA1.5-3B, Qwen2.5-VL-3B) on Jetson Orin Nano Super for offline assistive vision
- Benchmarked latency, memory footprint, and throughput across multiple VLM architectures and quantization schemes (q4f16_ft, q4), achieving stable on-device inference under 10–12 s latency at 640×480 resolution

3. Adaptive Graph Evidence Acquisition

May 2026 - Aug 2026

Independent Research@ByteDance, later supervised by Prof. He Bingsheng, National University of Singapore

- Developed an adaptive graph evidence-acquisition framework that separates cost-aware evidence construction from downstream classification and LLM-based fraud reasoning.
- Formulated evidence acquisition as a budget-constrained Markov decision process over composable graph tools, including neighbourhood expansion, Personalized PageRank, random-walk retrieval, community detection, and motif extraction.
- Investigated heuristic, value-selection, GRPO, and policy-learning acquisition strategies under explicit evidence and computation budgets.
- Built GCN, GraphSAGE, node2vec, PPR/RWR, motif-based, and policy-based baselines with multi-seed evaluation and controlled ablations.
- Studied sparse, adaptive, and relational fraud patterns under distribution shift, incomplete labels, and constrained evidence access.
- Developed evaluation protocols for evidence efficiency, structural faithfulness, downstream utility, and robustness to redundant or misleading graph evidence.

4. DistDNA

Aug 2026–Present

*Research Assistant under Dr Wu Zhaomin, supervised by Prof. He Bingsheng, National University of Singapore TBC***5. Graph Evidence Is Not Text Evidence**

Jun 2026–Present

Independent Research@ByteDance

- Proposed structural invariance as a necessary criterion for evaluating graph-grounded language models beyond task accuracy.
- Developed evaluations for serialization-order sensitivity, graph-isomorphism consistency, format stability, semantic interference, and distractor robustness.
- Decomposed structural unfaithfulness into syntactic instability, semantic interference, and distractor susceptibility.
- Demonstrated that output accuracy alone does not guarantee faithful use of graph structure.

6. Graph Grokking and Phantom Transitions

May 2026–Present

Independent Research@ByteDance

- Investigated delayed generalization, memorization basins, and abrupt performance transitions in graph neural networks and graph transformers.
- Studied optimization dynamics as an explanation for graph-model failures distinct from conventional oversmoothing and oversquashing.
- Analyzed the relationship between training trajectories, representation geometry, structural complexity, and late-stage generalization.

ACADEMIC SERVICE

Invited Reviewer, GroundLM Workshop at EMNLP 2026**Invited Reviewer**, VLM4RWD Workshop at NeurIPS 2026

OTHER EXPERIENCE

National University of Singapore, Lead Developer
2025

Jan 2025 - May

- Designed an autonomous mobile robot with ROS2, SLAM, and AMG8833 thermal imaging; navigated complex indoor environments and executed heat-targeted projectile firing with >90% directional accuracy
- Engineered multi-pass exploration logic (random walk → frontier-based → validation) and integrated A* pathfinding with real-time LiDAR-based obstacle avoidance, improving map coverage efficiency by ~35%

National University of Singapore, Robotics Group Leader
2025

Feb 2025 - Apr

- Developed a C++ serial communication interface on Raspberry Pi for teleoperation and claw actuation
- Synthesized infra-red sensing, ultrasonic braking, and servo-based multi-fingered claw control

NUS Astronomy Society, Member
Present

Oct 2024 -

- Planned and co-led AstroBash – a university-wide stargazing expedition to Langkawi, Malaysia
- Organized logistics and guided sky observation sessions for 30+ participants

NUS Photographic Society, Member

AWARDS & CERTIFICATES

- **Worldquant BRAIN Challenge** - Silver Medal, Worldquant
2025 *Feb*
- **2025 Mathematical Contest In Modelling** - Meritorious Winner, COMAP
2025 *May*
- **2026 Mathematical Contest In Modelling** - Honorable Mention, COMAP
2026 *May*
- **CDE Innovation & Research Award** — Merit Award
2026 *Jul*

SKILLS

- Programming & Data: Python, C/C++, Java, JavaScript, SQL, PySpark, Haskell
- Machine Learning: PyTorch, PyTorch Geometric, scikit-learn, Graph Neural Networks, Graph Transformers, Reinforcement Learning, PPO, Multi-Agent RL, Transformers, Vision-Language Models
- Graph Learning: GCN, GraphSAGE, Personalized PageRank, Random Walks, node2vec, Community Detection, Motif Analysis, Temporal Graph Analysis, Persistent Homology
- Computer Vision & Edge AI: OpenCV, YOLOv5, Ultralytics, PaddleOCR, Tesseract, TensorRT, NVIDIA Jetson Orin Nano, Model Quantization
- Systems & Deployment: FastAPI, Flask, Docker, Neo4j, Git, Linux, Streamlit, REST APIs, WebSockets
- Robotics & Embedded Systems: ROS2, SLAM, Raspberry Pi, Arduino, LiDAR, AMG8833 Thermal Sensor, Sensor Fusion
- Web & Interfaces: React, JavaScript Canvas, Tailwind CSS, Vite
- Research Areas: Graph Machine Learning, Graph-Grounded Language Models, Multi-Agent Systems, Fraud Detection, Edge AI, Geometric Deep Learning, Representation Learning
- Languages: Chinese (Native), English (Professional), Japanese (Intermediate), German (Basic)